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Total Number of Pages: 3

B.TECH
PET31103

3rd Semester Regular Examination 2016-17

SIGNAL & SYSTEMS

BRANCH(S): ECE, ETC

Time: 3 Hours

Max Marks: 100

Q.CODE: Y603

Answer Part-A which is compulsory and any four from Part-B.
The figures in the right hand margin indicate marks.

Part – A (Answer all the questions)

Q1 Answer the following questions: *dash fill up type* (2 x 10)

- a) All periodic signals are: ____ (energy/power/neither energy nor power)
- b) The even part of the signal $x(n) = e^{j\frac{\pi}{3}n}$ is ____.
- c) The Twiddle factor $W_N =$ ____.
- d) Autocorrelation function is ____ function.
- e) Check the linearity of $y(n) = Ax(n) + B$: ____
- f) For finite anti-causal sequence the ROC in Z-transform is ____.
- g) The normalized cross-correlation sequence $\rho_{xy}(l) =$ ____.
- h) The exponential form of Fourier series of a periodic signal $x(t)$ with period T is defined by ____.
- i) The system function described by the difference equation $y(n) = \frac{1}{2}y(n-1) + 2x(n)$ is ____.
- j) If the DFT of $x(n) = X(K)$, then $DFT\{x(n-m)_N\}$ is ____.

Q2 Answer the following questions: *Short answer type* (2 x 10)

- a) Describe all-zero system with an example.
- b) Describe elementary signals: unit sample sequence and ramp signal.
- c) Determine whether $x(n) = u(n)$ is energy or power signal, where $u(n)$ stands for unit step sequence.
- d) Determine whether $x(n) = \sin\left(\frac{6\pi}{7}n + 1\right)$ is periodic or not. If periodic find fundamental period.
- e) Define time-reversal property in Z-transform with its ROC.
- f) State Parseval's relation in Z-transform domain.
- g) Define zero padding and its importance.
- h) State and prove time shifting property of Fourier transform.
- i) Prove $[x_1(n) * x_2(n)] = [x_2(n) * x_1(n)]$, i.e. convolution for LTI system is commutative.
- j) Plot the discrete time sequence $y(n) = -2u(-n-2)$, where $u(n)$ is unit step sequence.

Part – B (Answer any four questions)

Q3 a) Draw the direct form-I and form-II realization of the LTI system given as $2y(n) + y(n-1) - 2y(n-3) = x(n-1) + x(n-3)$. **(10)**

b) State the relationship between convolution and correlation and prove it. **(5)**

Q4 a) Determine the response of the LTI system whose input $x(n]$ and impulse response $h(n]$ are given by, $x(n] = \left\{ \underset{\uparrow}{1}, 2, 3, 1 \right\}$ and $h(n] = \left\{ 1, \underset{\uparrow}{2}, 1, -1 \right\}$. Use graphical method to solve. **(10)**

b) Prove that for convolution of LTI system the distributive property holds. **(5)**

Q5 a) Derive convolution sum of LTI system and state its properties. **(10)**

b) Show that the necessary and sufficient condition for a relaxed LTI system to be BIBO stable is $\sum_{n=-\infty}^{\infty} |h(n)] \leq M_h < \infty$, for some constant M_h . **(5)**

Q6 a) Determine the response of LTI system governed by the difference equation, $y(n] - 0.5y(n-1] = x(n]$, for input $x(n] = (5)^n u(n]$, and initial condition $y(-1] = 2$. Use one-sided z-transform property. **(10)**

b) Find the Z-transform with its ROC for $n^2 u(n] - 3^n u(-n-1]$. **(5)**

Q7 a) Find the inverse Z-transform of the following **(10)**

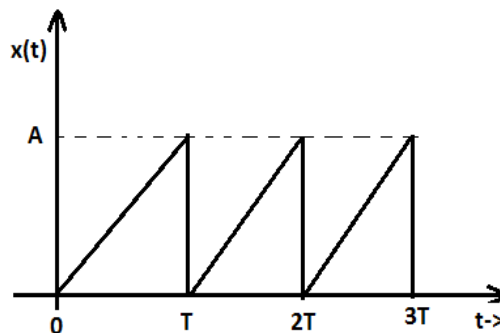
$$X(z) = \frac{1}{(1+z^{-1})(1-z^{-1})^2}$$

using partial-fraction expansion method.

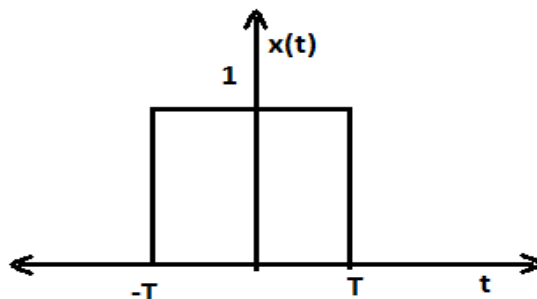
b) Compute 4-point DFT of the causal three sample sequence given by **(5)**

$$x(n] = \begin{cases} \frac{1}{3} & : 0 \leq n \leq 2 \\ 0 & : otherwise \end{cases}$$

Q8 a) Determine the Trigonometric form of Fourier series of the ramp signal shown below **(10)**



- b) Determine the Fourier transform of rectangular pulse shown below (5)



- Q9 a) Determine the total response $y(n), n \geq 0$ of the system described by the second order difference equation $y(n) - 2y(n-1) - 3y(n-2) = x(n) + 4x(n-1)$, when the input signal is $x(n) = 2^n u(n)$ and with initial condition $y(-2) = 0, y(-1) = 5$. (10)
- b) Consider the following two sequences $x_1(n) = \{2, 3, 1, 4\}$ and $x_2(n) = \{5, 2, 1\}$. Find the circular convolution of $x_1(n)$ and $x_2(n)$. (5)